

First-Token Broadcasters: Mechanistic Analysis of Language Identity and Distributed Robustness in Transformers

Anonymous ACL submission

Abstract

How is language identity represented and causally maintained in transformers, and to what extent do the mechanisms identified in one language set transfer to other languages and training regimes? We introduce **Language Identity Head Ablation (LIHA)**, a causal intervention that zeros each attention head individually and measures the resulting language switch rate. Across our multilingual evaluation sets, LIHA identifies heads whose ablation has disproportionate effects on output language. In GPT-2, a small set of *first-token broadcaster* heads—led by L6H1 (switch rate 0.32, 3.23σ above the population mean)—attend persistently to the first prompt token across generation steps. The combination of persistent first-token attention and causal sensitivity suggests that these heads participate in maintaining output language. Following ablation, we observe statistically significant redistribution of first-token attention ($p < 10^{-5}$) toward heads in higher layers for the ablations tested; this pattern is consistent with downstream compensation, although attention redistribution alone does not establish a feedforward causal cascade. We also compare Qwen2.5-1.5B-Base and Qwen2.5-1.5B-Instruct, which are matched in architecture and parameter count but differ in training history. The base model is nearly flat (max SR=0.016, 200/336 heads at SR=0.0), whereas the instruct model concentrates causal influence at layer 0, led by L0H5 (SR=0.224, 8.93σ above mean). Finally, GPT-2 heads identified from the European-language experiments show no observed switch effects on the tested Chinese and Russian prompts, while other early-layer heads do show effects. Because script, typology, and tokenization vary together in this comparison, these results do not isolate a specifically script-driven mechanism.

1 Introduction

Whether foundation models represent language identity as a single language-universal signal or

as script- and family-specific circuitry is a question with direct typological stakes: a mechanism identified on one language family may simply not exist, in any recoverable form, for a typologically distant one. In GPT-2, a small set of attention heads attend strongly to the first prompt token across generation steps. For prompts whose first token is language-informative, these heads show persistent first-token attention. We call these *first-token broadcaster* heads.

Language confusion has been documented behaviorally across multilingual models (Marchisio et al., 2024), while related work has studied code-switching limitations and language-dependent internal representations (Wendler et al., 2024). Prior mechanistic work has also identified language-specific neurons and shown that manipulating them can influence generation language (Tang et al., 2024). We take a complementary approach at the attention-head level: for each attention head, we zero its output during generation and directly measure how often the output language changes. This causal lens reveals not just which heads matter, but how language identity is organized across heads and how that organization changes across training regimes.

Our main findings: (1) partial localization in GPT-2, led by L6H1 (SR=0.32, 3.23σ above mean; §4); (2) first-token broadcasting, with top heads attending to the first prompt token at mean weight 0.85–0.94 every step (§5); (3) hierarchical compensatory redistribution ($p < 10^{-5}$), recruiting heads *above* the ablated layer in all three ablated heads tested (§4); (4) convergent probing evidence at the same layers (§5); (5) instruction tuning reorganizes the circuit toward a sharp layer-0 concentration in Qwen2.5-1.5B-Instruct ($14\times$ higher max SR, 8.93σ) versus a near-flat base model (§6); (6) the European-language broadcaster heads identified in GPT-2 do not show switch effects on the tested Chinese or Russian prompts, which instead show

effects in other early-layer heads (layers 0–4) (§6).

Contributions. (a) We introduce LIHA as a head-level causal intervention for measuring changes in output language following attention-head ablation; (b) we identify GPT-2 heads whose causal influence coincides with persistent attention to the first prompt token; (c) we characterize first-token attention redistribution following ablation and test its layerwise directionality; (d) we compare matched Qwen2.5-1.5B base and instruct variants and find stronger localization of causal effects in the instruct model; and (e) we show that European-language broadcaster heads identified in GPT-2 do not generalize unchanged to the tested Chinese and Russian prompts. The BLOOM analysis is an exploratory cross-architecture comparison because only every third layer is sampled.

2 Related Work

Elhage et al. (2021) established the residual stream framework used throughout this work; Wang et al. (2022) traced a 26-head circuit for indirect object identification in GPT-2; Conmy et al. (2023) introduced automated circuit discovery; and Meng et al. (2022) localized factual associations to MLP modules. On the multilingual side, Conneau et al. (2020) showed multilingual BERT acquires cross-lingual structure without explicit supervision, and Marchisio et al. (2024) document that state-of-the-art models frequently switch to unintended languages; we approach this mechanistically via causal ablation rather than behavioral analysis. Tang et al. (2024) identified language-specific neurons via activation analysis—a language-*selective* signal that need not coincide with the language-*causal* heads LIHA isolates. On the typological side, Pires et al. (2019) showed multilingual BERT’s cross-lingual transfer degrades with typological distance between languages, and Chi et al. (2020) located universal grammatical-relation structure within its representations; LIHA complements this line of work by asking not whether cross-lingual structure exists, but which specific components causally determine *which* language identity is broadcast. Linear probing (Alain and Bengio, 2017; Tenney et al., 2019) provides convergent evidence (§5), and inference-time steering (Zou et al., 2023; Li et al., 2023) motivates our amplification experiments (§4).

3 Experimental Setup

Models and implementation. GPT-2 small (Radford et al., 2019) has 12 layers, 12 heads, and 144 total heads. Qwen2.5-1.5B (Yang et al., 2024) has 28 layers, GQA (2 KV heads, 12 Q heads), and 336 total heads; we compare its base and instruct variants, matched in architecture and parameter count but differing in training history. For cross-architecture replication, we additionally evaluate BLOOM-1b7 (BigScience Workshop et al., 2022), a model deliberately pretrained on 46 natural languages; it has 24 layers, 16 heads per layer (hidden dim 1024, head dim 64), and 384 total heads. Each head’s output is isolated by zeroing its post-projection contribution to the residual stream (Elhage et al., 2021), implemented via PyTorch forward hooks on `o_proj`. We use `langdetect` (Nakatani, 2010) for language classification and generate 40 tokens per prompt with greedy decoding.

Dataset. We construct a multilingual evaluation set drawn from FLORES-200 (NLLB Team et al., 2022), spanning five European languages (English, French, German, Spanish, Italian; 500 sentences each for GPT-2 and 25 sentences per language for Qwen. Chinese and Russian use 100 sentences each in the extended experiments. The 125 Qwen prompts are a subset of the 2,500 European-language prompts rather than additional prompts. Prompts are filtered for quality and verified via 3-way majority vote (`langdetect`, `langid` (Lui and Baldwin, 2012), `fastText` (Joulin et al., 2016)); all European languages achieved $\geq 99.6\%$ verified prompts.

Formal definition of LIHA. Let $\mathcal{M}$ be a transformer with L layers and H heads per layer. For head $h_{l,i}$, define the ablated model $\mathcal{M}^{-h_{l,i}}$ as $\mathcal{M}$ with that head’s residual stream contribution set to zero. The **switch rate** over prompt set $\mathcal{X} = \{x_1, \dots, x_n\}$ is:

$$S(h_{l,i}) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}[\mathcal{L}(\mathcal{M}^{-h_{l,i}}(x_j)) \neq \mathcal{L}(\mathcal{M}(x_j))] \quad (1)$$

A head with high $S(h_{l,i})$ has greater causal influence on detected output-language changes under this intervention. This measure does not imply that the head uniquely encodes language identity or that it specifically promotes the target language rather than suppressing alternatives.

Protocols. We sweep all 144 GPT-2 heads and all 336 Qwen2.5-1.5B heads (both variants); for BLOOM we sweep every third layer (128 heads). Switch rates report 95% bootstrap CIs (10,000 samples); attention redistribution following ablation uses 100,000-sample permutation tests on first-token attention weight deltas. Zero-ablation is a commonly used intervention in mechanistic interpretability (Elhage et al., 2021; Wang et al., 2022); we discuss mean ablation as an alternative in §8.

Compute and reproducibility. All experiments were conducted on a system equipped with an NVIDIA GeForce RTX 3060 GPU using Python 3.12 and PyTorch. Package dependencies are specified in the accompanying repository’s `requirements.txt`. Experiments cover seven languages: English, French, German, Spanish, Italian, Chinese, and Russian. Generation uses greedy decoding (`do_sample=False`), making generation deterministic for a fixed model and input. Randomized statistical procedures, including bootstrap and permutation analyses, use random seed 42; the probing classifier likewise uses `random_state=42`. We additionally set `DetectorFactory.seed=0` to make `langdetect` deterministic. The accompanying repository provides the code, dependency specification, and experiment scripts required to reproduce the reported results.

4 Results: Single-Head Ablation and Distributed Robustness (GPT-2)

Figure 1 shows the language switch rate across all 144 GPT-2 heads, sparse with elevated rates in a cluster spanning layers 0–10. The mean switch rate is 0.163 ($\sigma=0.049$). L6H1 leads at SR=0.32 (99.3rd percentile, 3.23σ above mean), the highest-ranked head by both switch rate and effect size; a second tier (L0H4, L3H1, L9H9) sits at 0.28, with 95% bootstrap CIs overlapping L6H1’s (Table 1), so we treat L6H1 as the leading candidate rather than statistically separated. No language-switch events were observed for English under the tested ablations (baseline accuracy 99.2%, unchanged under ablation); Italian is most vulnerable (switch rate 0.6 under L6H1 ablation; Appendix A.3).

Figure 1(b) shows accuracy under progressive multi-head ablation on the full 2,500-prompt European set. Baseline accuracy is 43.5% [0.416, 0.454]. Ablating L6H1 alone reduces accuracy to 39.2% [0.373, 0.411]; ablating the top ten heads

Table 1: **Top five GPT-2 heads by switch rate** with 95% bootstrap CIs (500 prompts per language).

Head	Switch Rate	95% CI
L6H1	0.32	[0.16, 0.52]
L0H4	0.28	[0.12, 0.44]
L3H1	0.28	[0.12, 0.48]
L9H9	0.28	[0.12, 0.48]
L7H3	0.24	[0.08, 0.40]
Population mean	0.163	–

reduces it to 32.4% [0.306, 0.343], an 11.1-point total reduction. Critically, accuracy **never falls below chance (20%)** and degradation is monotonic, which is consistent with redundant or distributed encoding; the result does not by itself establish a specific circuit structure.

Attention redistribution following ablation.

The top-5 mean first-token attention delta exceeds the permutation null for all three ablated heads ($p < 10^{-5}$); the largest increases cluster *above* the ablated layer (layer-clustering $p < 0.006$; full table in Appendix A.2). Ablating L0H4 (layer 0) recruits layer 2; ablating L6H1 (layer 6) recruits layers 7–9; ablating L9H9 (layer 9) recruits layers 10–11. The pattern is compatible with downstream compensation, but it does not by itself establish a feedforward causal cascade. Amplifying individual top heads by $2 \times -5 \times$ produces no observed accuracy improvement at the tested scales. Thus, simple amplification of an individual head was insufficient to recover language accuracy in this experiment; this result is motivated by prior inference-time representation interventions, but does not directly test the intervention proposed in Li et al. (2023).

5 Results: First-Token Broadcasting and Probing

Figure 2 (Appendix B.1) shows L6H1’s attention on correct and confused prompts: every query position attends to the first token at weight 0.62–1.00. Broadcaster heads attend to the first prompt token with mean weight 0.85–0.89 across generation steps, with stable low entropy across all 40 steps (mean 1.11 vs. 1.34 for random heads; Appendix B.2). Confused prompts show *higher* first-token attention in L6H1 (0.923 vs. 0.847 on correct prompts): confusion does not simply coincide with loss of first-token access, suggesting that downstream processing may determine the effective use of this information. A descriptive taxonomy of

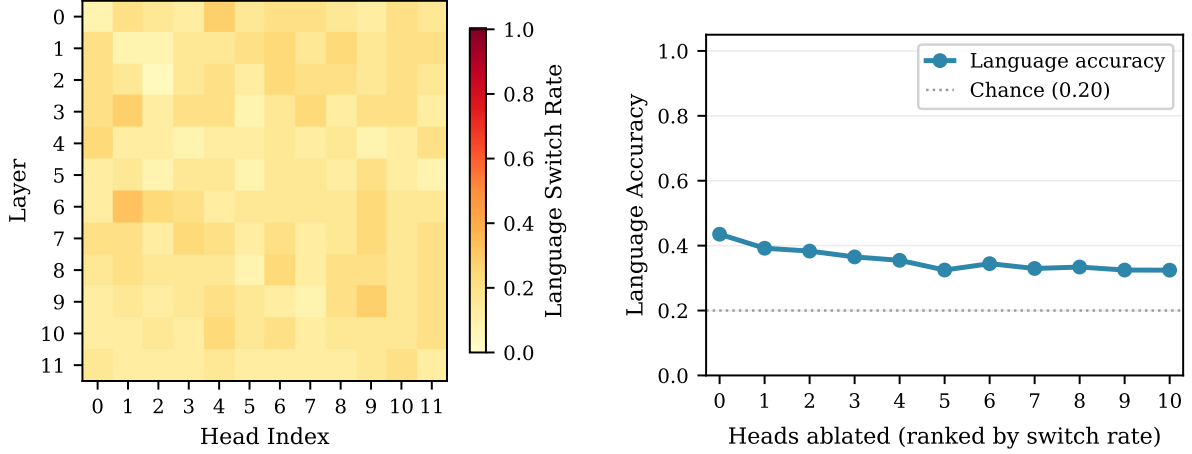


Figure 1: **GPT-2 head-level language identity and distributed robustness.** (a) Language switch rate across all 144 GPT-2 attention heads. A small cluster spanning layers 0–10 shows disproportionate causal influence. L6H1 leads at 0.32 (99.3rd percentile, 3.23σ above mean); a second tier (L0H4, L3H1, L9H9) sits at 0.28. (b) Language accuracy under progressive multi-head ablation (2,500-prompt evaluation, 95% bootstrap CIs). Monotonic gradual degradation—without reaching chance in the tested sequence—is consistent with redundant or distributed encoding. Dotted line: chance (0.20).

heads (broadcaster vs. distributed) is given in Appendix B.3.

In GPT-2, probing accuracy rises from 36% at the embedding layer to 96% at layers 10–11; in BLOOM, it reaches 100% at layers 18–24. Layers containing top causal heads show substantially higher probing accuracy in both models (GPT-2: 85% vs. 58%; BLOOM: top-switch layers all above 92%), providing convergent evidence that language information is decodable in layers containing larger causal effects, although probing alone does not establish that the probed feature is causally responsible.

6 Results: Training Regime and Language-Family Specificity

We apply LIHA to Qwen2.5-1.5B-Base and Qwen2.5-1.5B-Instruct—matched in architecture and parameter count, with the same GQA configuration but having different training histories—sweeping all 336 heads with 125 prompts. The **base model** is nearly flat: mean SR=0.0036 ($\sigma=0.0046$), maximum SR=0.016 (L0H0), and 200/336 heads at SR=0.0, indicating that, under single-head ablation and this sample size, causal effects on detected output language are not strongly localized to any individual head. The **instruct model** is qualitatively different: L0H5 leads at SR=0.224 [0.152, 0.296], 8.93σ above the population mean—a $14\times$ increase in max SR over

the base model. All 12 layer-0 heads are elevated (layer-0 mean SR=0.093 vs. 0.023 for layers 1–27); six heads exceed SR=0.1 (Table 7). No language-switch events were observed for English under the tested ablations in either variant; German is most vulnerable (switch rate up to 0.44), replicating the language-specificity pattern observed in GPT-2.

Cross-model comparison. Table 2 summarizes all models. The progression from GPT-2 (distributed, middle layers) to Qwen-Base (near-uniform) to Qwen-Instruct (layer-0 concentrated) is consistent with a reorganization associated with the post-training regime. Because the Qwen comparison contains only one base/instruct pair and the variants differ in training history beyond a single isolated intervention, it does not by itself establish instruction tuning as the causal ingredient.

Extending to Chinese (zh) and Russian (ru), the top European heads (L6H1, L0H4, L3H1, L9H9; switch rates 0.28–0.32) produce SR=0.0 on both—the European-language broadcaster heads show no observed switch effects on the tested Chinese and Russian prompts. Different heads, concentrated in layers 0–4, govern non-European languages instead—the same broad early-layer locus as the instruction-tuned Qwen model (full breakdown in Appendix C). Notably, L1H10, a *distributed* head, retains a non-trivial switch rate on both Chinese and Russian (0.20) where the European broadcasters show no observed switch effects, suggesting that

Table 2: Cross-model comparison of language identity organization.

Property	GPT-2	BLOOM-1b7	Qwen-1.5B Base	Qwen-1.5B Instruct
Heads swept	144	128 [†]	336	336
Max SR	0.32	0.16	0.016	0.224
Top head σ	3.23	2.60	2.70	8.93
# heads SR>0.1	15	4	0	6
Critical layers	0–10	0–21	none	0
English switch events	none	none	none	none

[†]Every 3rd layer sampled; 128 of 384 heads.

some distributed heads may have broader cross-lingual effects than the localized heads in this experiment.

7 Discussion

A training-regime hypothesis. Pretrained-only GPT-2 distributes the language signal across 10+ middle-layer heads, while BLOOM shows similarly broad effects in the sampled layers. Qwen-Instruct instead shows stronger layer-0 concentration than the matched Qwen-Base model. This pattern is consistent with an association between the post-training regime and circuit localization, but the single base/instruct pair does not isolate instruction tuning as the causal ingredient.

The early-layer convergence. Chinese and Russian in GPT-2 recruit heads concentrated in layers 0–4; Qwen-Instruct concentrates tightly at layer 0 specifically; and the Qwen base model’s single strongest head is also at layer 0. These early layers sit closest to the tokenizer, where tokenization and prompt formatting can expose language-correlated information early. Models that resolve language identity early may therefore show effects in this broad locus, although the present experiments do not establish a common causal origin, and the GPT-2 non-Latin-script effect is diffused across a four-layer window rather than sharply localized to a single head the way the Qwen-Instruct effect is.

Implications for inference-time intervention. Ablating ten heads causes only 11.1 points of monotonic, never-at-chance accuracy decline, hierarchically absorbed downstream. Qwen-Instruct’s layer-0 concentration raises the possibility that some instruction-tuned models may be more amenable to targeted intervention, but our results do not establish greater intervention reliability. More broadly, the observed variation across GPT-2, Qwen-Base, and Qwen-Instruct indicates that a single-head in-

tervention may not transfer across training regimes or language sets.

Future directions. Path patching (Wang et al., 2022) would distinguish heads that promote the target language from those that suppress alternatives. Whether the layer-0 concentration in Qwen-Instruct extends to larger instruction-tuned models, and whether it enables more reliable language control, are open questions.

Overall interpretation. Our results support a narrower conclusion than a single universal language-identity circuit. In GPT-2, several heads have measurable causal influence on output language, and a subset combine this influence with persistent attention to the first prompt token. Localization and effect magnitude vary across models, training regimes, and language sets. These observations are consistent with language-related computation being implemented by partially distributed, model-dependent circuits rather than a single universal head-level mechanism.

8 Limitations

GPT-2 switch rates use 500 prompts per European language; Qwen uses 125, useful for the base-vs.-instruct comparison but with a minimum resolvable nonzero switch rate of $1/125 \approx 0.008$, so the large number of Qwen-Base heads at SR=0.0 (200/336) partly reflects measurement resolution. Zero-ablation cannot distinguish heads promoting the correct language from those suppressing alternatives; we compared this intervention with mean ablation as a sensitivity analysis, which showed only a weak (though statistically significant) negative rank correlation with zero-ablation rankings (Spearman $\rho = -0.24$, $p < 0.01$)—full analysis in Appendix E. Qwen2.5-1.5B’s Grouped Query Attention means zeroing a Q-head slice has different internal semantics than in standard MHA, since multiple Q-heads share KV projections. Our inter-

vention zeros the selected post-projection Q-head output slice; the implications of this intervention for GQA-specific circuit semantics require further validation. The Qwen comparison controls for architecture and scale but spans a single model family; broader replication across instruction-tuned families is needed to generalize the training-regime hypothesis. Finally, our Chinese/Russian comparison confounds *script* (non-Latin orthography) with broader *typological* distance (morphology, word order, syntactic structure): both languages differ from the European set on both dimensions simultaneously, so we cannot yet attribute the early-layer shift to script identity specifically versus typological distance more generally. Disentangling these would require testing a Latin-script language that is typologically distant from the European set (e.g., Vietnamese) alongside a non-Latin-script language that is typologically close to it (e.g., Serbian in Cyrillic), which we leave to future work.

Responsible use. This work analyzes publicly available pretrained models with constructed prompts; no personal data is involved. **Positive impact:** understanding first-token broadcasting could inform more reliable multilingual systems and targeted mitigations for language-switching failures. **Potential negative impact:** causal head-localization methods could also be used to alter a model’s language behavior in unintended ways, and the layer-0 concentration observed in the instruction-tuned model may make some interventions easier to target. **Mitigation:** we report aggregate statistics rather than a ready-to-use intervention procedure, and the same localization that may improve understanding of a mechanism can also support monitoring and hardening. **Acknowledgement:** Omitted for anonymous review.

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A GPT-2 Ablation, Redundancy, and Compensation

A.1 Full GPT-2 Ablation Sweep

Table 3: All GPT-2 heads with switch rate > 0.15 .

Head	Switch Rate	Delta Acc.
L6H1	0.32	+0.04
L0H4	0.28	+0.04
L9H9	0.28	-0.12
L3H1	0.28	+0.00
L10H4	0.24	-0.08
L9H11	0.24	-0.04
L8H6	0.24	+0.00
L9H4	0.24	+0.00
L7H3	0.24	-0.04
L1H0	0.24	+0.08
L1H10	0.24	+0.04
L5H1	0.24	+0.00
L6H8	0.24	-0.04
L3H4	0.20	-0.04
L4H11	0.20	+0.00

A.2 Attention Redistribution Following Ablation

Table 4: **Heads with the largest increases in first-token attention when L6H1 is ablated.** These heads show positive attention deltas; the set clusters in layers 7–9, above the ablated layer 6 ($p=0.002$). Because many heads are tested and the displayed heads are selected by effect size, we do not interpret the uncorrected per-head p -values as confirmatory evidence.

Head	Base	Ablated	Δ	z
L9H8	0.267	0.275	+0.007	3.45**
L7H3	0.745	0.751	+0.006	3.01**
L7H10	0.873	0.879	+0.006	2.70**
L7H1	0.833	0.838	+0.005	2.65**
L7H6	0.702	0.705	+0.003	2.10*

** $p<0.01$; * $p<0.05$; permutation test, $n=100,000$.

A.3 Per-Language Switch Rates

Table 5: Per-language switch rates for the top five GPT-2 heads from Table 1, computed on the 5 hand-written sentence-starter prompts per language (§3).

Head	EN	FR	DE	ES	IT
L6H1	0.0	0.2	0.2	0.4	0.6
L0H4	0.0	0.2	0.2	0.4	0.4
L3H1	0.0	0.2	0.2	0.4	0.4
L9H9	0.0	0.2	0.0	0.4	0.6
L7H3	0.0	0.2	0.2	0.2	0.4

Note: this qualitative breakdown uses the 5 hand-written prompts per language and therefore differs slightly from the aggregate switch rates over the full 500-prompt set reported in Table 1.

B First-Token Broadcasting Diagnostics

B.1 Attention Patterns on Correct and Confused Prompts

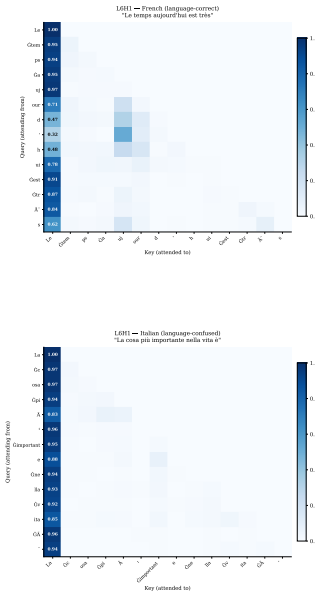


Figure 2: **L6H1 attention on a correct French prompt (left) and confused Italian prompt (right).** First-token focus is consistent in both cases; confusion arises downstream.

B.2 Entropy Over Generation Time

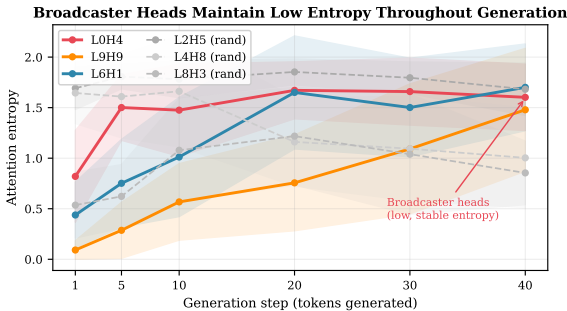


Figure 3: **Attention entropy over generation steps.** Broadcaster heads (solid) maintain lower entropy (mean 1.11) than random heads (dashed, mean 1.34) across all 40 steps.

B.3 Functional Taxonomy of Language-Critical Heads

Table 6: Descriptive taxonomy of top language-critical heads.

Head	Type	First-Tok Attn	Entropy
L6H1	Broadcaster	0.876	0.517
L10H4	Broadcaster	0.848	0.677
L7H3	Broadcaster	0.892	0.538
L1H10	Distributed	0.112	2.013

C Training-Regime and Language-Family Specificity

C.1 Top Qwen2.5-1.5B-Instruct Heads

Table 7: Top five Qwen2.5-1.5B-Instruct heads by switch rate (125 prompts, 95% bootstrap CIs). No language-switch events were observed for English under the tested single-head ablations.

Head	SR	95% CI	σ above mean
L0H5	0.224	[0.152, 0.296]	8.93
L0H11	0.144	[0.088, 0.208]	5.33
L1H9	0.136	[0.080, 0.200]	4.97
L0H7	0.120	[0.064, 0.176]	4.25
L1H7	0.112	[0.064, 0.168]	3.89
Population mean	0.026	—	—

C.2 European vs. Chinese/Russian Head Specificity

Table 8: European-language versus Chinese/Russian head effects. Left: top European-language heads show no observed switch events on the tested Chinese/Russian prompts. Right: other heads show nonzero switch rates, concentrated in layers 0–4. These small-sample results are descriptive and do not isolate script from typology or tokenization.

Head	EU	ZH	RU	Head	All	ZH	RU
L6H1	0.32	0.00	0.00	L0H0	0.40	0.20	0.60
L0H4	0.28	0.00	0.00	L4H5	0.30	0.40	0.20
L3H1	0.28	0.00	0.00	L4H2	0.30	0.40	0.20
L9H9	0.28	0.00	0.00	L0H1	0.20	0.00	0.40
L1H10	0.24	0.20	0.20	L0H7	0.20	0.20	0.20

All denotes the mean switch rate averaged across the Chinese and Russian prompt sets combined.

C.3 Extended Language Prompts

Table 9: Chinese and Russian prompts used in the extended language experiments. Russian is shown in transliteration; Cyrillic originals are in data/prompts_ru.csv.

Lang	Prompt
zh	今天的天气非常
zh	我想告诉你关于
zh	科学家们发现了
zh	生活中最重要的事是
zh	从前有一个
ru	<i>Segodnya pogoda ochen'</i>
ru	<i>Ya khotel by rasskazat' vam o</i>
ru	<i>Uchyonye obnaruzhili, chto</i>
ru	<i>Samoye vazhnoye v zhizni — eto</i>
ru	<i>Odnazhdy zhil-by</i>

D Statistical Test Details

Switch rate CIs use 10,000 bootstrap samples (resampling with replacement, 2.5th/97.5th percentiles). Attention redistribution following ablation uses 100,000-sample permutation tests: the observed top-5 mean delta is compared against the distribution of top-5 means from 100,000 random draws of size 5 from the full 143-head delta population. Layer-clustering tests compare observed layer entropy of the five largest attention-increase heads against the entropy from 100,000 random 5-head draws.

E Mean-Ablation Methodological Check

We additionally attempted mean ablation—replacing each head’s output with its mean activation across the dataset—as a methodological check. The resulting switch rates showed only a weak negative rank correlation with zero-ablation rankings (Spearman $\rho = -0.24$, $n = 144$ heads), statistically distinguishable from zero at $p < 0.01$ under the standard large-sample approximation for Spearman’s ρ . We treat this as a sensitivity analysis rather than evidence that either intervention is universally preferable. The sign and magnitude are consistent with the two interventions perturbing different aspects of the representation, but the present analysis does not establish why the rankings differ. In particular, we do not use this result to confirm that zero-ablation is uniquely appropriate; zero-ablation is retained as the primary intervention because it provides a direct removal of the head’s post-projection residual contribution.